

GradeLine
User Guide

Existing-ground long sections and driveway design from survey DXF data

Document	GradeLine User Guide
Applies to	GradeLine 1.1.0: Windows desktop app, single HTML file and website
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Project repository	https://github.com/TReno06/GradeLine
Website	https://treno06.github.io/GradeLine/

Intended use. GradeLine is a screening and visualisation aid. It does not replace engineering design, a council's own requirements, or professional sign-off. Section 13 sets out its limits.

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What's new in GradeLine 1.1

- **Breaklines**: 3D kerb and gutter linework in the DXF is built into the surface, so a kerb keeps its face (Section 4.2).
- **AS/NZS 2890.1:2004**: domestic-driveway grade rules, a footpath crossfall check, and the Standard's Figure C1 ground clearance templates, with B85 as the default for houses (Sections 7.4 and 8).
- **Crossfall as a cross-section**: square across the two alignments at any design chainage (Section 9).
- **Layers out of position** are flagged when a DXF loads (Section 3.2).
- **A new sample job**: a curved street, kerb with a layback, a sloping lot and a bending driveway with two edge alignments (Section 2.2).
- Arc and closed alignments, a 3D orientation cube, **Reverse direction**, editable vehicle dimensions, a resizable split view, collapsible menus, and light and dark themes.

1. Introduction

GradeLine reads survey data from a DXF file, builds a triangulated ground surface (a TIN), cuts a long section along one or more alignments, and lets the user lay a straight-grade driveway design over that section and check it against AS/NZS 2890.1:2004. It runs entirely on the user's own computer. No survey data is uploaded, and the application works offline.

1.1 Purpose

This guide explains how to load survey data, prepare it, read the resulting views, design and check a driveway, read the crossfall between two alignments, save work, and export results to AutoCAD or PDF. It is written for civil and survey practitioners who are comfortable with chainages, reduced levels (RL) and long sections.

1.2 Capabilities

- Build a 3D surface from POINT and 3DFACE survey shots and contour polylines, with 3D kerb and gutter linework built in as breaklines.
- Flag layers that sit away from the rest of the survey before they are used.
- Read the existing-ground long section along any alignment found in the DXF, including arcs, with grade breaks, grade percentages, a regular chainage ladder and a manually set datum line.
- Lay out a proposed driveway profile, including a kerb and a layback template, and check it against the AS/NZS 2890.1:2004 domestic driveway limits.
- Test whether a B85 or B99 design vehicle would scrape, using the Standard's Figure C1 templates.
- Read the crossfall square across two alignments, as a cross-section at any design chainage.
- Save a complete job and reopen it later; export to DXF, CSV and PDF.

1.3 What GradeLine is not

GradeLine is not a general CAD or road-design package. The surface follows breaklines only where the DXF supplies them as 3D linework. The driveway tool is a screening aid over a design the user authors. Other proposed-surface work remains in the user's CAD or design software.

2. Getting GradeLine and first run

GradeLine is available in two forms, plus a website for trying it. All are the same application, and a saved job opens in any of them.

Form	How to obtain and start it	Notes
Windows desktop app	Download `GradeLine_1.1.0_x64-setup.exe` from the latest release on the repository and run it.	The only way to install GradeLine as an app. Installs for the current user only; no administrator rights are needed. The installer is not code-signed, so Windows may display "Windows protected your PC": choose **More info**, then **Run anyway**. Opens `.gradeline` job files by double-click.
Single HTML file	Download `GradeLine.html` from the latest release and open it in Chrome or Edge.	One self-contained file: no installation, no server, and no network access needed.
Website	Open the hosted application address in a browser.	For trying GradeLine without downloading anything. It is not installable; use the desktop app for that.

2.1 First run

1. Open GradeLine. The sidebar is on the left and the main view is on the right (Figure 1). Every sidebar section starts collapsed; click a heading to open it. Sections you open are remembered next time.
2. On the empty start screen in the main view, select **Load sample driveway job**. (To return to it later, select **Clear loaded data** first.)
3. Drag in the 3D view to orbit and scroll to zoom. Hold Shift and drag to pan.
4. Open **Display** and select the **Existing terrain**, **Driveway** and **Crossfall** tabs in turn.
5. To work with real data, select **Clear loaded data**, then **Upload DXF** (Section 3).

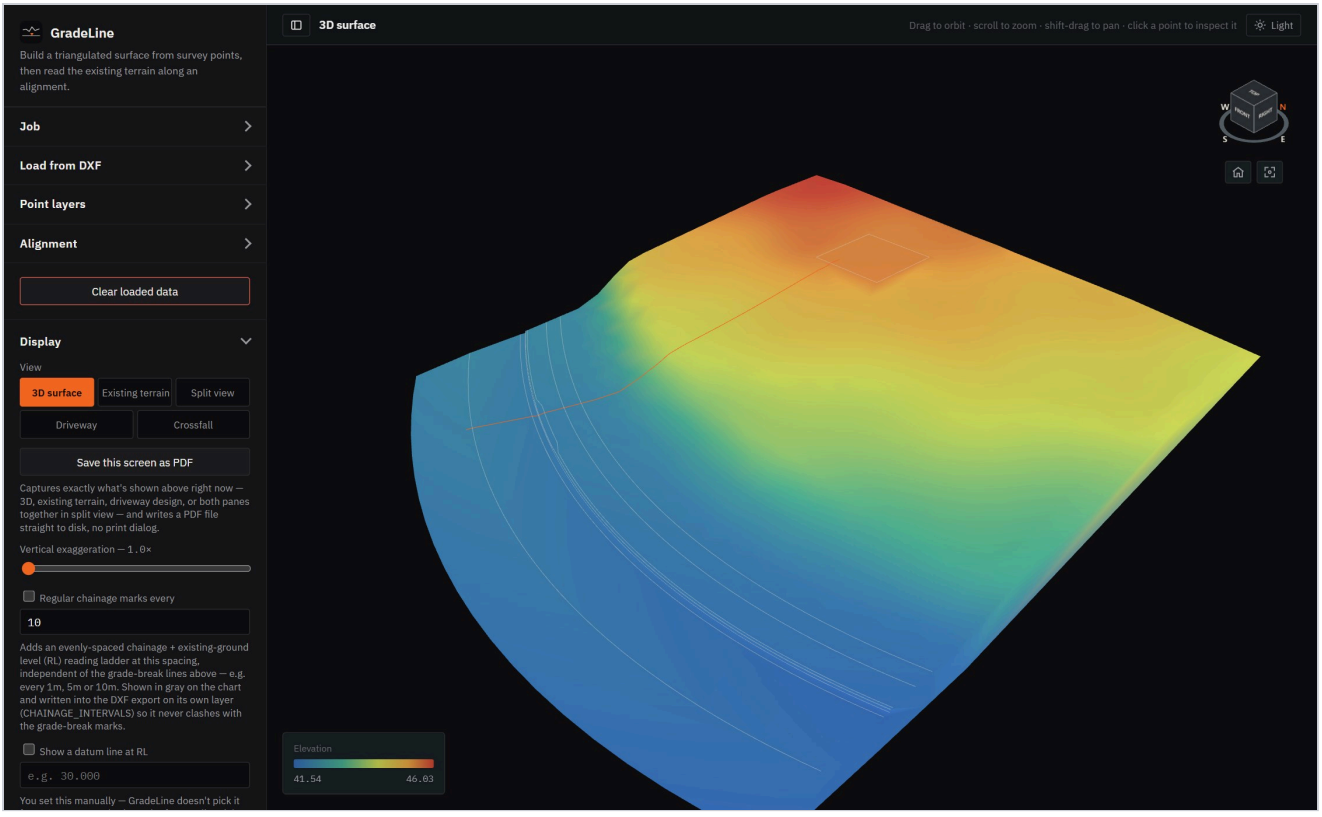


Figure 1. GradeLine with the sample driveway job loaded, on the 3D surface tab. The sidebar holds the job, DXF, layer, alignment and display controls; the view tabs (3D surface, Existing terrain, Split view, Driveway, Crossfall) are in the Display section. The kerb, gutter and footpath strings are drawn on the surface as faint lines.

2.2 The sample job

The sample is a made-up residential lot laid out like a real job, so every feature has something to show:

- A street that curves, with a crowned road, a 150 mm barrier kerb and gutter, and a verge with a concrete footpath. The kerb and gutter are supplied as 3D breaklines.
- An existing layback across the driveway: a 30 mm lip at the gutter invert rising 70 mm over 450 mm, with splays back to full kerb height.
- A lot that climbs a front bank to a garage slab and also falls to the east.
- A 3 m driveway that leaves the kerb square to it and bends into the garage, with one alignment along each edge. Each edge has its own driveway design that passes every default check.

3. Loading survey data

DXF is the only survey format GradeLine reads. Export the survey from the CAD or survey package as a DXF, then select **Upload DXF** in the **Load from DXF** section. There is no need to strip the drawing down first; unwanted layers can be excluded after loading.

3.1 Entities read

DXF entity	Use in GradeLine
POINT	A survey shot (X, Y, Z), used to build the surface.
3DFACE	Face corners are used as survey shots. Corners shared between faces are counted once.
3D LINE, ARC, and polylines whose level changes along them	Breaklines: the surface's triangles are made to follow them (Section 4.2). An ARC has a single elevation in DXF, so it takes its levels from the surveyed points lying on it.
Polyline at one constant elevation	A contour line; its vertices are added to the surface as points.
Flat polyline (no elevation)	Offered as an alignment for the long section. Arcs (bulges) and closed polylines are followed exactly.

3.2 Point layers

When the DXF has more than one layer, a **Point layers** checklist appears with All and None buttons. Each row shows the number of points, and the number of breaklines on that layer, for example `KERB_TOP (24 + 42 lines)`. Only ticked layers feed the surface, and a ticked layer contributes both its points and its lines. With several layers, none is ticked at first, so that the choice is made deliberately for each load. Tick only genuine survey layers; leave out text labels, blocks and title-block linework.

Layers out of position. GradeLine groups layers by where they sit in plan and treats the group around the alignments as the site. Any other layer is marked with a ⚠ note giving its rough distance and direction, for example ⚠ `~54m W of the rest — check position`, and is listed in the load summary. Typical causes are a surface moved aside in CAD, or title-block linework near coordinate (0, 0). Ticking such a layer stretches the surface across the gap. The note never unticks anything; it is a prompt to check.

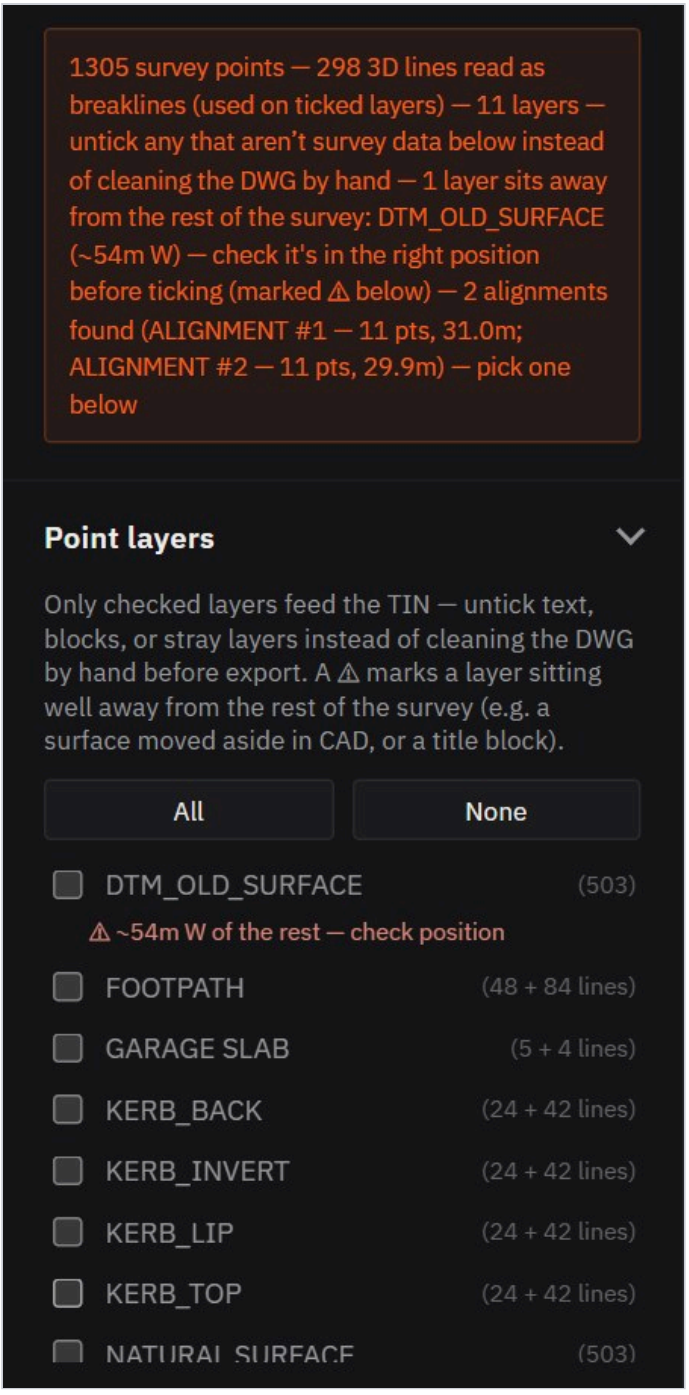


Figure 2. The load summary and Point layers checklist for a DXF in which one surface layer (DTM_OLD_SURFACE) sits about 54 m west of the rest of the survey. Row counts show points plus breaklines.

3.3 Alignments

Flat polylines on a layer whose name contains "alignment" are preferred as alignment candidates. If no such layer exists, every flat polyline is offered. The **Alignment** dropdown lists them as Alignment 1, Alignment 2 and so on, with layer, point count and length. The selection can be changed at any time without reloading.

Reverse direction flips which end of the alignment is chainage 0. Any driveway design on it is mirrored so it stays on the same ground, and the stricter property-line grade limit moves to the road end.

After loading, the surface rebuilds automatically. For large files a progress overlay with a Cancel button appears; cancelling returns the application to a clean, empty state.

4. Preparing the data

Unsuitable points produce an unreliable surface. Review the data before relying on a section. All filters apply immediately; the file does not need to be reloaded.

4.1 Filters

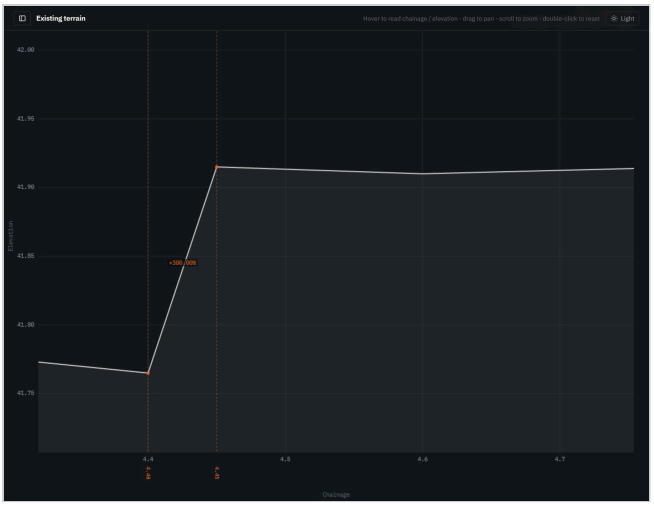
Control	Default	Purpose
Point layers checklist	Nothing ticked when several layers exist	Exclude layers that are not survey data (Section 3.2).
Use breaklines (Display section)	On	Build the ticked layers' 3D linework into the surface. Untick to compare with a plain triangulation.
Ignore points with elevation (Z) = 0	On	Removes DXF text-label rows, which always report Z = 0.000 and would otherwise pull the surface to zero.
Ignore points beyond a coordinate limit	Off (limit 0)	Drops stray outlier clusters. Set a maximum |X| or |Y| from the survey origin and tick the box.

4.2 Breaklines

A plain triangulation joins survey points with the shortest triangles it can find, which can run across a kerb from the gutter to the top of kerb and turn a vertical face into a ramp. With breaklines, the triangles are forced to follow the 3D linework, so the kerb face, gutter invert and lip appear in the surface and in every long section. Figure 3 shows the same section across a 150 mm barrier kerb with and without breaklines.

Below the **Use breaklines** checkbox (Display section), a notice reports how many breakline segments were built into the surface, and any that were skipped because they cross another breakline or could not be fitted. **Show breaklines on surface**, just below it, draws them on the 3D view as faint lines.

With breaklines



Without breaklines

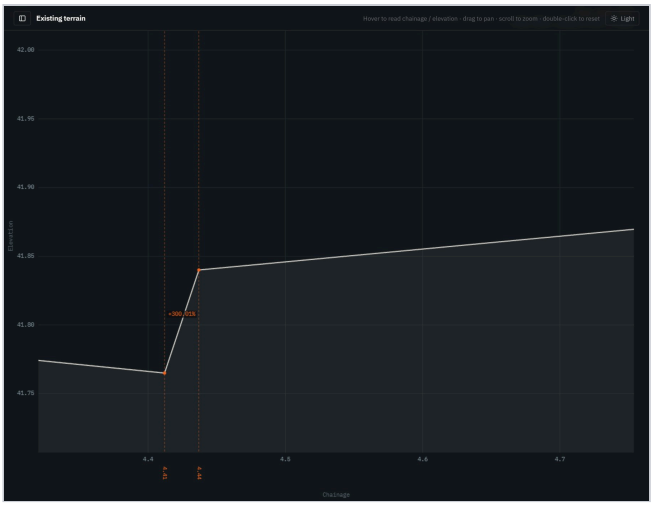


Figure 3. A long section across the sample's barrier kerb, zoomed to about 0.4 m. With breaklines the section drops to the gutter invert (RL 41.765) and rises the full 150 mm face to the top of kerb. Without them, a triangle cuts the corner: only about 75 mm of the face shows and the top of kerb becomes a slope.

4.3 Summary and diagnostics

The **Summary** section reports the point count, triangle count, elevation range and alignment length. The **Diagnostics** section is updated after every rebuild and reports:

- **Disconnected clusters**: groups of points that the surface has bridged although they are unrelated, often a stray survey from another site.
- **Near-duplicate points**: points within 0.01 units of each other in X and Y (detection is approximate, using a 0.01 grid).
- **Unusually steep triangles**: possible errors in Z, or genuine kerbs, walls and cliffs.

> **Note** A diagnostic flag is a prompt to inspect the data, not proof of an error. A kerb face or retaining wall will trigger the steep-triangle check.

Clear loaded data removes all data and returns to the starting state.

5. The views and layout

Five tabs in the Display section select the main view. All draw from the same loaded data.

Tab	Content
3D surface	The triangulated ground in 3D, coloured by elevation, with an elevation legend.
Existing terrain	The long section along the active alignment (Section 6).
Split view	The 3D surface above the long section, with the driveway tools alongside.
Driveway	The long section with the driveway design controls (Sections 7 and 8).

Tab	Content
Crossfall	Two alignments and the cross-section between them (Section 9).

5.1 3D surface

- **Navigation**: drag to orbit, scroll to zoom, Shift-drag to pan.
- **Orientation cube** (top right): click a face, edge or corner to view from that direction; the ring shows north, south, east and west. The buttons below it return to the home view and zoom to the extents of the data.
- **Vertical exaggeration**: a slider from 1x to 10x. It starts at 1x (true scale). Raise it to make gentle terrain readable, and return to 1x before judging real gradients.
- **Display options**: show triangulation edges, survey points, the alignment and the breaklines. The alignment is draped along the surface; any part off the surveyed area is drawn dashed.
- **Point inspection**: click (without dragging) on the surface to read the real-world X, Y and Z of that location.

5.2 Split view

Hovering along the long section places an orange marker at the matching position on the 3D surface. Drag the divider between the two panes to resize them. The driveway tools panel on the right can be shown or hidden with **Driveway tools** in the top bar, and widened by dragging its edge.

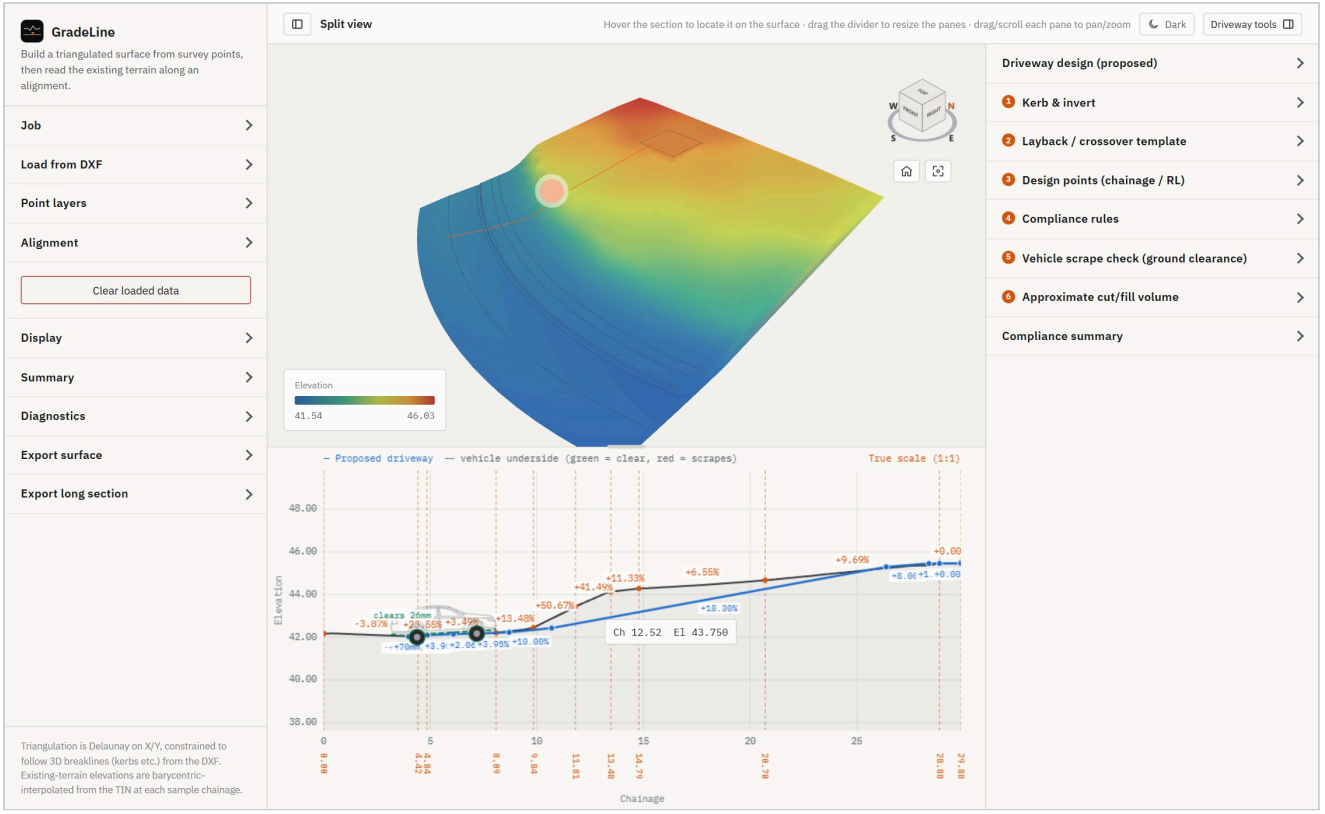


Figure 4. Split view in the light theme. Hovering the long section (chainage 12.52) places the orange marker on the 3D surface; the orientation cube and home and zoom-extents buttons are at the top right of the 3D pane.

5.3 Layout and themes

- Every sidebar and tool-panel section folds shut by clicking its heading. All start collapsed, and the ones you open are remembered.
- The button at the top left of the main view hides or shows the whole sidebar.
- **Light** / **Dark** in the top bar switches the theme; the choice is remembered. PDFs capture whichever theme is showing.

5.4 Saving the current screen as a PDF

Save this screen as PDF (Display section) captures whichever tab is showing, both panes in split view, and writes a PDF directly to disk, with a header, date and the current alignment name.

6. Reading the long section

The **Existing terrain** tab plots the ground profile along the active alignment, with chainage on the horizontal axis and RL on the vertical axis (Figure 5). When the alignment has a driveway design, it is drawn over the ground in blue.

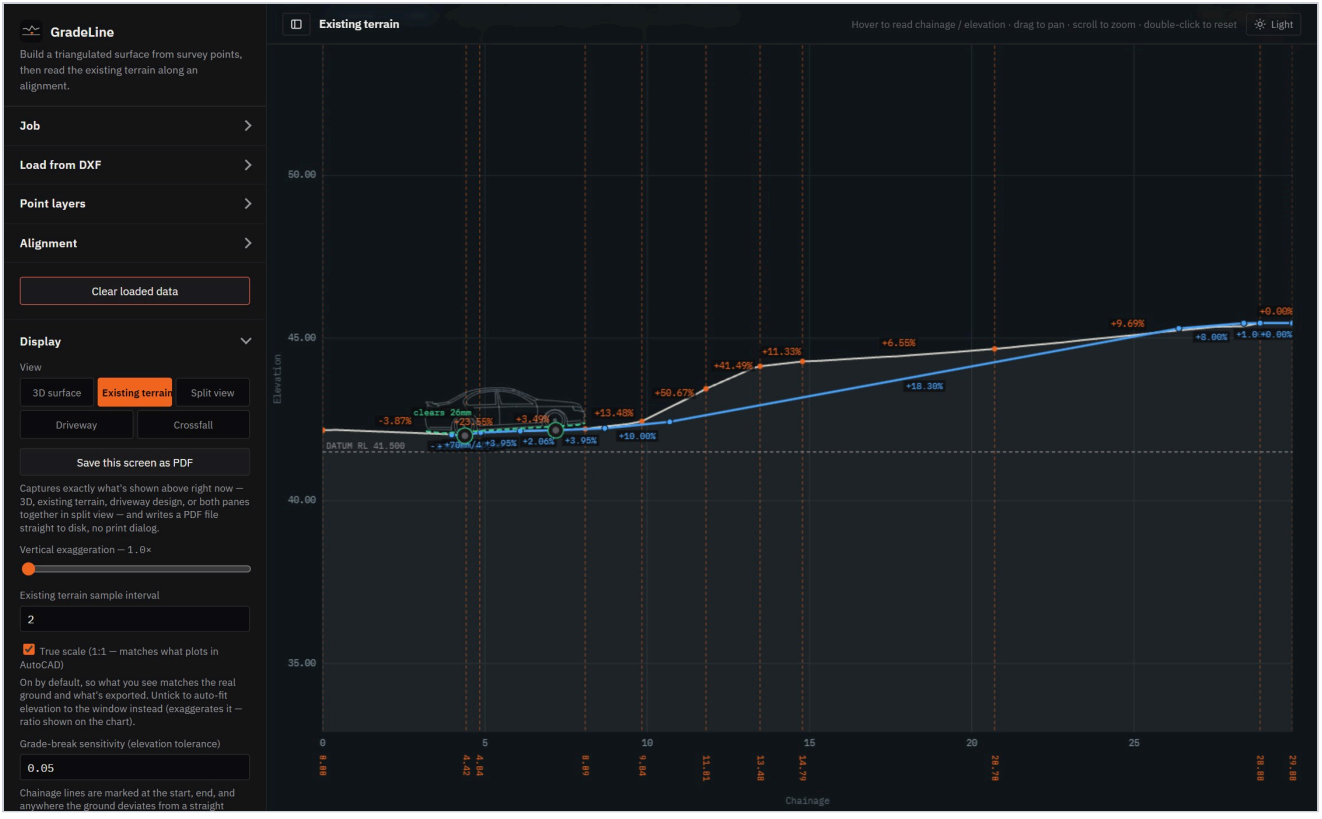


Figure 5. The Existing terrain tab on the sample job (Alignment 1, the driveway's west edge): grade-break chainage lines and signed grade percentages for the ground (grey), the driveway design (blue) with the vehicle check, and a datum line at RL 41.500.

6.1 Scale and navigation

- **True scale (1:1)** is on by default so the chart matches what plots in AutoCAD. Real sections usually look nearly flat at true scale. Untick it to fit the elevation range to the window; the chart then prints

its vertical exaggeration so the distortion is never hidden.

- **Pan and zoom**: drag to pan, scroll to zoom toward the cursor, double-click to reset. The axis labels gain decimals as you zoom in.
- Hover over the chart to read chainage and elevation.

6.2 Annotations and controls

Control	Default	Effect
Existing terrain sample interval	2	Adds extra points between the real breaks. It does not change the shape, minimum or maximum of the section, which are found exactly from the surface.
Grade-break sensitivity (elevation tolerance)	0.05	Chainage lines are drawn at the start, the end, and wherever the ground departs from a straight grade by more than this tolerance.
Regular chainage marks every	Off; spacing 10	Adds an evenly spaced chainage and RL reading ladder (grey), independent of the grade breaks.
Show a datum line at RL	Off	Draws a dashed line at an RL you enter, labelled "DATUM RL". GradeLine never chooses the datum. It is always exported, even when it lies outside the visible levels.

Between each pair of grade breaks the chart shows the signed grade as a percentage. Sample interval, grade-break sensitivity and the chainage ladder are three separate controls: sensitivity decides where grade breaks fall, the ladder uses only the spacing entered, and the sample interval moves no marks.

7. Driveway design

The **Driveway** tab lays a proposed driveway over the existing-ground section and checks it. The model is straight legs between design points, with no vertical curves. It screens a design you author; it does not generate one. The chart is at the left and the numbered tool sections are at the right (Figure 6).

Open **Driveway design (proposed)** and tick **Design a proposed driveway profile** to begin. Every alignment keeps its own independent design: switching the alignment switches which design is edited.

7.1 Kerb and invert (section 1)

Enter the **Kerb invert chainage**, the **Kerb invert RL** (gutter level) and the **Kerb reveal (mm)**, the rise from the invert to the top of kerb or layback lip, then select **Set kerb points**. GradeLine inserts two design points at the same chainage, forming a vertical face. The fields start blank because kerb details differ between councils.

7.2 Layback and crossover template (section 2)

Two rise-over-run legs, in millimetres, straddle the kerb invert as on a council layback drawing: the road-side leg runs back from the invert, the property-side leg forward from the top of kerb. The fields are pre-filled with a typical layback (road side 25 mm over 450 mm; property side 70 mm over 450 mm), which should be checked against the council detail. Select **Apply layback template** to insert the points.

> **Note** The kerb face and both layback legs are never checked against the grade limits. They are shown as a rise and run in millimetres. This is deliberate.

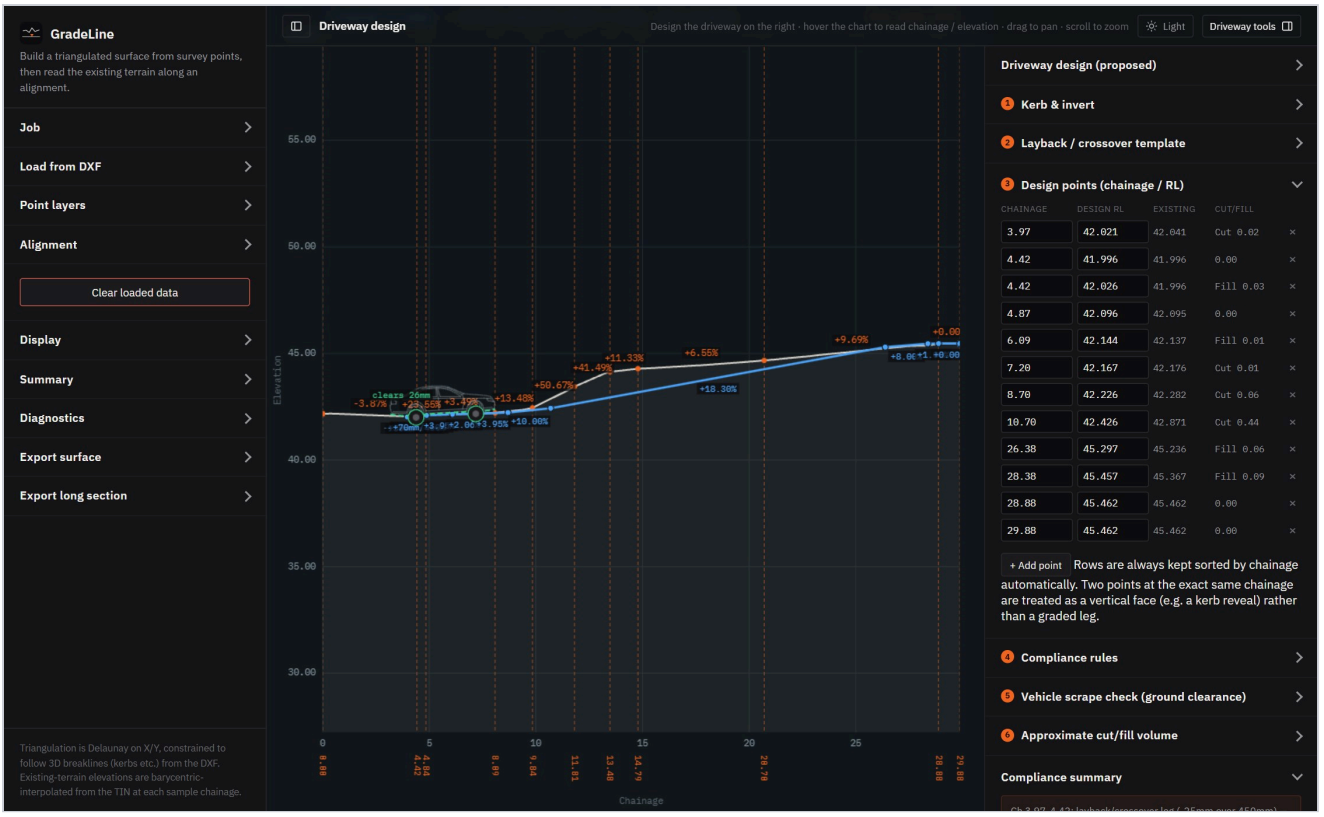


Figure 6. The Driveway tab on the sample job. The design points table (section 3) shows the layback and kerb rows at chainages 3.97 to 4.87 (the two rows at 4.42 form the 30 mm layback lip), 4% up the verge, 2% across the footpath (6.09 to 7.20), the transitions and the main grade into the garage.

7.3 Design points (section 3)

When design is first enabled, a design point is created at each end of the alignment, matching existing ground. The table lists chainage, design RL, the existing RL and the cut or fill at each point. Values can be edited, points added with **+ Add point** and removed with the cross at the row end. Rows are always sorted by chainage. Two points at exactly the same chainage form a vertical face.

7.4 Compliance rules (section 4)

The thresholds default to AS/NZS 2890.1:2004 for **domestic driveways**, and every value can be changed.

Check	Default	Source
Maximum grade within the property	25% (1 in 4)	Cl 2.6.2
Stricter limit on the first leg, kerb to property line	5% (1 in 20), optional	Cl 2.6.2
Grade across the footpath	2.5% (1 in 40) or flatter over at least 1.0 m, optional	Cl 3.3(d), via Cl 2.6.2
Grade change needing a transition: summit	12.5% (1 in 8)	Cl 2.5.3(d)
Grade change needing a transition: sag	15% (1 in 6.7)	Cl 2.5.3(d)
Minimum transition length	2.0 m	Cl 2.5.3(e)

GradeLine does not know where the footpath is, so the footpath check looks along the crossing, from the kerb to where the driveway first climbs steeper than the property-line limit, for one continuous stretch at 2.5% or flatter. A grade change is a summit when the grade falls in the direction of travel and a sag when it rises. The **Compliance summary** lists every leg and check; failing legs are drawn in red on the chart. Councils often set their own footpath and crossing requirements: check them before relying on a pass.

7.5 Approximate cut and fill (section 6)

Enter a **Corridor width (m)** and GradeLine reports cut, fill and net volume in cubic metres along the design's chainage span: the corridor width multiplied by the integrated difference between design and existing levels. It is indicative only: it takes no account of crossfall, batters or earthworks shaping, and it is not a surface-to-surface volume.

8. Vehicle scrape check

Section 5 of the Driveway panel tests whether a design vehicle would ground out along the proposed profile, using the method of AS/NZS 2890.1:2004 Appendix C: the vehicle's underside template is rested on the profile at both wheel contact points and rotated to the local grade, then swept along the driveway. Tick **Check a design vehicle's underside against this profile** to enable it.

Template (Figure C1)	Wheelbase	Front / rear overhang	Use
B85 (default)	2800 mm	920 / 1190 mm	Domestic driveways only.
B99	3050 mm	950 / 1200 mm	All other driveways.

Both templates have 120 mm ground clearance between the axles. At each end the underside follows a 10° line drawn from the tyre contact point (the approach angle at the front, the departure angle at

the rear), so it stays at 120 mm until about 0.68 m out from the axle and then rises to the bumper.

Control	Function
Design vehicle	B85 (domestic driveways) or B99 (all other driveways).
Vehicle reversing (rear leads)	By default the vehicle faces toward increasing chainage, matching a chart read from street to garage. Tick to check it backing down the driveway; the result differs because the overhangs differ.
Vehicle dimensions	Wheelbase, overhangs, ground clearance and the approach and departure angles, editable per design and saved with it, for example to match vehicle tracking software. An angle of 0° gives a flat overhang. Readouts say "custom vehicle dimensions" once any value is changed; **Reset to defaults** restores the Figure C1 values.
Position slider	Sweeps the vehicle along the driveway.
Jump to worst point	Finds the tightest clearance anywhere on the design and moves the vehicle there.

Case 1: the vehicle clears. On the sample job the B85 clears the whole driveway. Its tightest point is on the layback at chainage 3.69, where the readout reports `B85 clears by 26mm at its tightest point (ch 3.69)`. The dashed underside line is green.

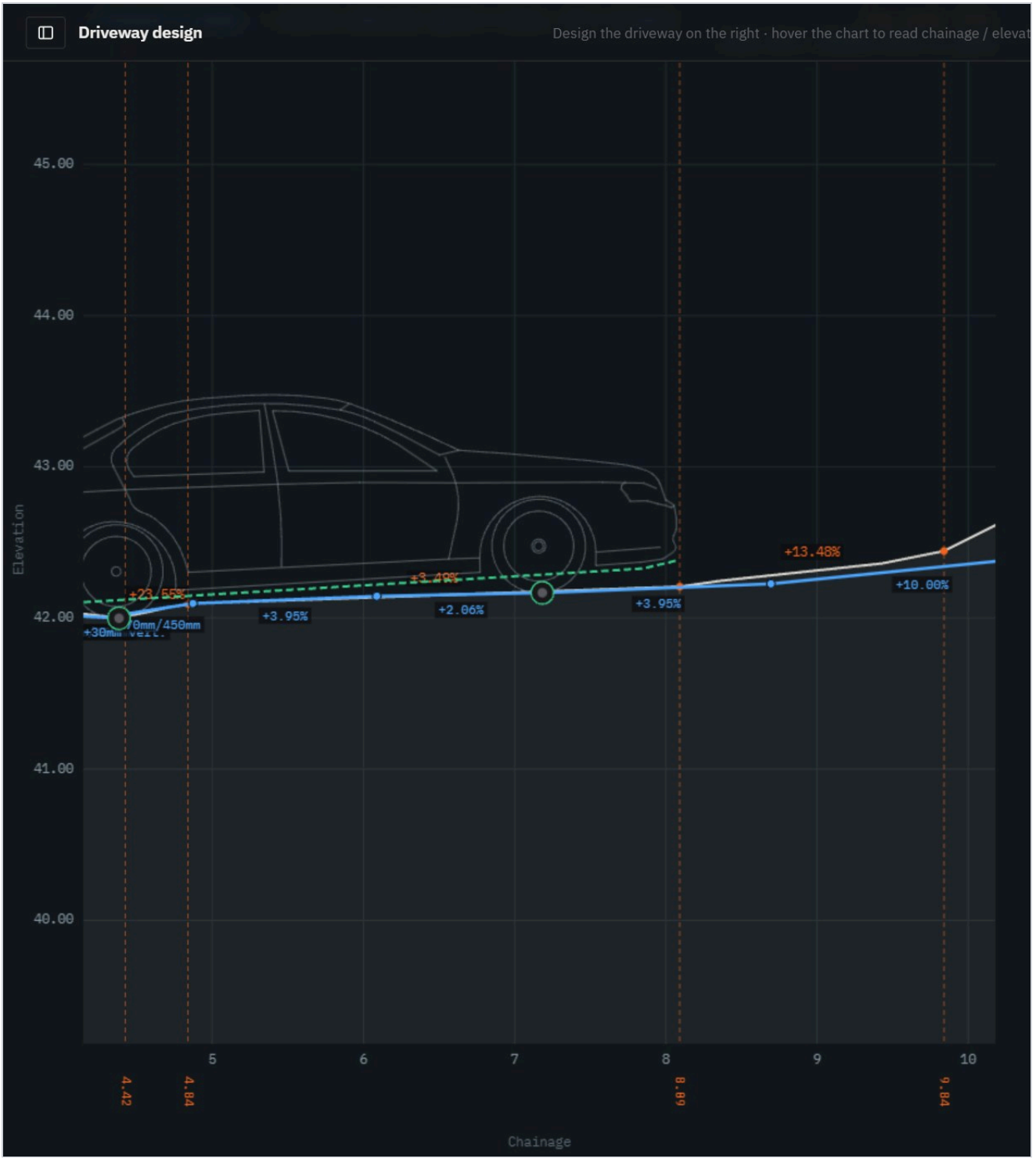


Figure 7. The B85 at its tightest point on the sample driveway, crossing the layback and footpath. The green dashed underside line and wheel markers show that it clears. Grey is existing ground, blue is the design.

Case 2: the vehicle scrapes. Raising the design point at chainage 26.38 by 350 mm creates a sharp crest: 20.5% up, then 9.5% down, a 30% summit change. The 2 m legs either side satisfy the transition-length rule, but the B85 grounds on the crest: the dashed line turns red and the readout reports `B85 SCRAPES by 89mm (worst point at ch 26.38)`. This is why the Standard asks for the template check on grade changes of 18% or more.



Figure 8. The B85 straddling the crest at chainage 26.38. The red dashed underside line shows the vehicle grounding; the label reads "SCRAPES 89mm".

> **Note** At true 1:1 scale a few tens of millimetres are almost invisible beside tens of metres of chainage. Rely on the readout and the Compliance summary, which always includes the worst point for the selected direction, or scroll to zoom in as in Figures 7 and 8.

9. Crossfall

The **Crossfall** tab shows the fall across a driveway between two alignments, typically its two edges. It needs at least two alignments. Where an alignment has no driveway design enabled, existing ground is used for that side.

9.1 Procedure

1. In **Crossfall between two driveway designs**, select Alignment A and Alignment B. Two different alignments are chosen by default.
2. In **Chainage readout**, pick one of Alignment A's design chainages from **Design chainage**, type any chainage, or drag the orange marker across the two profile charts.
3. Read the cross-section at the bottom of the view: a straight line from A's level to B's level, with the fall in millimetres and percent and the width across (Figure 9).

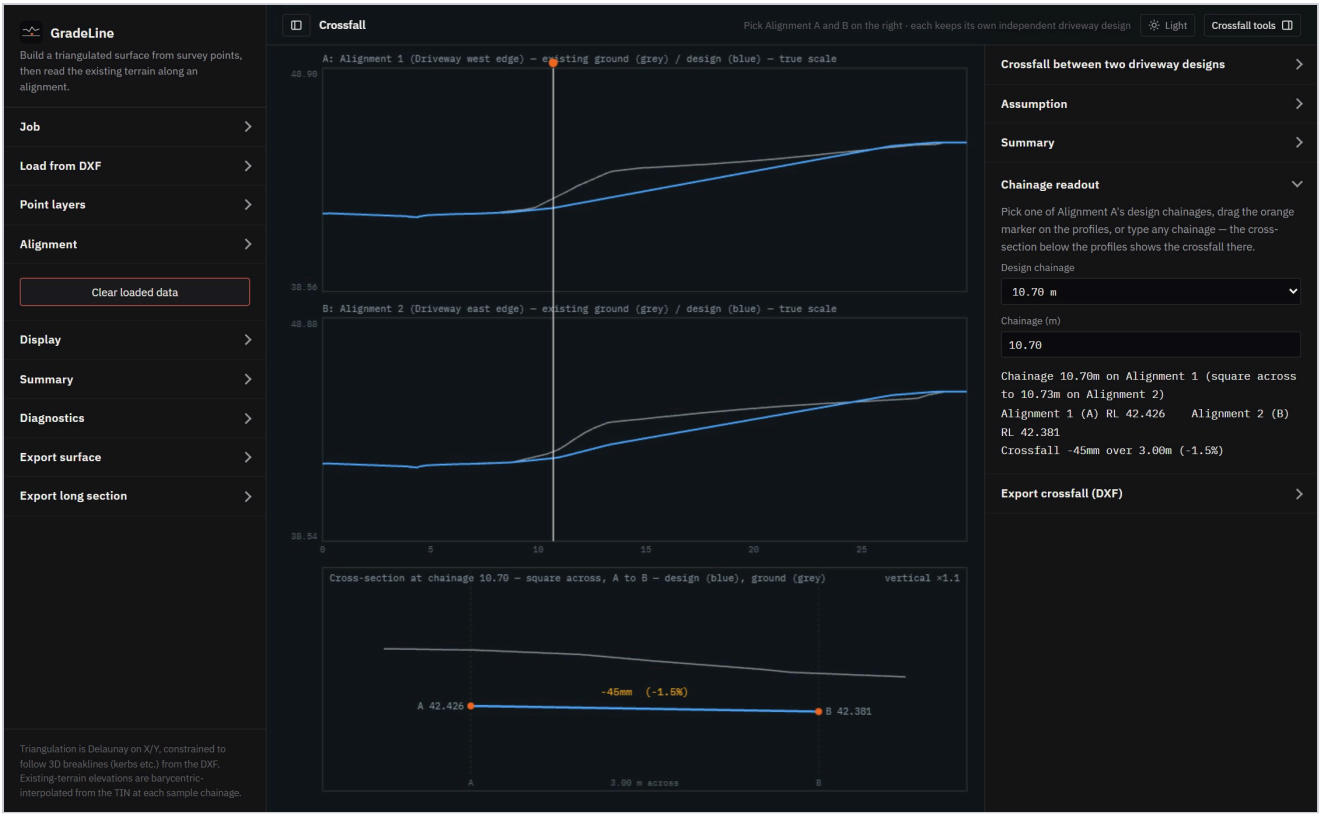


Figure 9. The Crossfall tab on the sample job at design chainage 10.70. The two profiles are at the top with the marker; the cross-section below is square across from the west edge (A, RL 42.426) to the east edge (B, RL 42.381): a fall of 45 mm over 3.00 m (−1.5%), with existing ground along the same line in grey.

9.2 Comparison methods

Method	How the two sides are paired	Suitable when
Square across the alignments (default)	At each chainage on A, a line is drawn square to A and B is read where it crosses. Past B's ends, the nearest point on B is used.	Always; required on a bend, where the inside edge is shorter and the same chainage value on each edge is not the same place across the driveway.
Matching chainage value	Chainage 10.00 on A is compared with chainage 10.00 on B.	Only when both alignments start at a comparable point and run parallel.

On the sample job the driveway bends: the west edge is about 1.1 m shorter than the east edge, and matching chainage values would report a false crossfall of more than 150 mm on the bend.

9.3 Reading results

The **Summary** gives the worst crossfall along the whole driveway and its chainage. The readout gives the chainage on each alignment, both RLs, and the fall over the width. The cross-section's vertical scale is true where it fits and otherwise exaggerated, with the factor shown at the top right. The two profile charts are true scale.

> **Note** At a kerb chainage the two kerb faces rarely line up exactly square across, so one side may read its invert and the other its lip. Read the crossfall just either side of the kerb.

10. Saving and reopening jobs

A job is a `.gradeline` file. It holds the loaded survey data including breaklines, the chosen alignment, every alignment's driveway design and vehicle dimensions, the crossfall setup, the filter settings, the datum and the view settings. The file is compressed. The surface itself is rebuilt on opening, exactly as after a fresh DXF load.

- **Save**, **Save As** and **Open** (in the Job section) use the operating system's own file dialogs, so any folder can be chosen. Save writes back to the last file used; Save As selects a new one.
- If a save fails (for example the drive is disconnected, permission is lost or the disk is full), GradeLine reports it. Use Save As to write elsewhere.
- Browsers without file-picker support (such as Firefox and Safari) download the file on Save and ask for the file on Open.
- In the desktop app, double-clicking a `.gradeline` file opens it. The first time, Windows shows a "Pick an app" dialog: choose GradeLine and select **Always**. Saving after a double-click opens Save As.

GradeLine also remembers the sidebar's view, filter and datum settings, the theme, and which sections are open, and restores them at the next visit even when no job is opened. This never includes survey or design data.

11. Exporting results

GradeLine exports the long section, the surface and the crossfall view, and can save the current screen as a PDF. Long-section files are named after the alignment and content, for example `Alignment 1 Both.dxf`.

Export (button)	Format	Contents
Download existing terrain (CSV)	CSV	The existing-terrain profile as chainage and elevation (`Alignment N Existing.csv`).
Existing / Proposed / Both (Export long section)	DXF	Existing ground with grade breaks, grade percentages, the optional chainage ladder and datum; with a driveway design, the proposed design and the section table.
Download TIN surface (DXF)	DXF	The triangulated ground as 3DFACE entities in true X, Y, Z.
Existing / Proposed / Both (Export crossfall)	DXF	The two alignments' profiles and the crossfall along the driveway, with a data table. Proposed and Both are disabled until at least one alignment has a driveway design on.
Save this screen as PDF	PDF	An image of the current tab, written directly to disk.

The desktop app presents a native Save As dialog for every export.

11.1 Contents of the long-section DXF

All geometry is true scale in metres: X is chainage and Y is true elevation, with no vertical exaggeration. Apply the required plot scale in AutoCAD. Each kind of content is on its own layer.

Layer	Contents
LONG_SECTION	Existing-ground polyline (white).
CHAINAGE_MARKS	Chainage lines and values at the start, the end and each grade break.
GRADE_LABELS	Grade percentage between breaks.
CHAINAGE_INTERVALS	The regular chainage and RL ladder, if enabled.
DATUM	The datum line and its labels, if set.
PROPOSED_DRIVEWAY	The design polyline (blue), design-point and leg labels, and any failing checks.
SECTION_TABLE	The CAD-style table of distance, grade, existing level, proposed level and chainage.

Labels that would overlap are stacked onto additional rows rather than omitted, and the table columns are spaced apart with leader lines back to their true chainage.

11.2 Opening the DXF in AutoCAD

1. In AutoCAD, choose File, then Open, set the file type filter to DXF, and select the file.
2. To save a DWG, use Save As and choose the DWG file type.
3. To bring the geometry into an existing drawing, open the DXF, copy the geometry, and paste it with Paste to Original Coordinates. The DXFIN command works only on a blank drawing.

> **Note** GradeLine writes DXF only, in a deliberately plain form with no version header and with the section lines written as old-style polylines. This is the form that opens reliably in AutoCAD. GradeLine cannot write DWG directly; convert the DXF in AutoCAD or with a converter.

12. Troubleshooting

Symptom	Likely cause and action
"Fewer than 3 valid points remain" straight after loading a DXF with several layers	No layer is ticked yet (Section 3.2). Open Point layers and tick the survey layers.
A layer is marked ⚠ away from the rest of the survey	It sits well away from the site in plan: a surface moved aside in CAD, or title-block linework. Leave it unticked, and check its position with the surveyor if it should be part of the site.
The surface has spikes down to zero	Text-label rows with Z = 0 are included. Keep Ignore points with elevation (Z) = 0 ticked and untick label layers.
A raised bridge or blob far from the site	A stray cluster. Check Diagnostics for disconnected clusters, then untick its layer or set a coordinate limit.
A kerb looks like a ramp in the long section	Its 3D linework is not in use: tick the kerb layers and Use breaklines. If the kerb was surveyed as points only, with no lines, there is nothing to build breaklines from.
The notice says some breaklines were skipped	They cross another breakline. Check the linework in CAD; the rest of the surface is unaffected.
No alignment appears in the dropdown	The DXF contains no flat polyline. Place the alignment on a layer named like "alignment" as a flat polyline and export again.
A section of the menu seems to be missing	It is collapsed. Click its heading to open it.
The long section looks almost flat	This is correct at true 1:1 scale. Zoom in, or untick True scale and read the vertical exaggeration label.
The vehicle appears to clear but the readout reports a scrape	The gap is too small to see at true scale. Rely on the readout, or zoom in.

Symptom	Likely cause and action
The crossfall looks wrong on a bend	Check that the method is Square across the alignments (Section 9.2).
An export does nothing in the desktop app	Complete the native Save As dialog; check that it is not hidden behind another window.
Buttons stop responding or the application is slow	Save the job, reload, and reopen the job file.

12.1 Good practice

- Save a `.gradeline` job as soon as the data has been filtered, and again after design changes.
- Review the layer notes and Diagnostics after every load, before relying on a section.
- Judge gradients at true scale and 1x vertical exaggeration; use exaggeration only to see detail.
- Confirm the AS/NZS 2890.1 limits against the local council's requirements before relying on a pass.

13. Limitations and assumptions

- **Breaklines come only from 3D linework.** Points surveyed along a kerb with no lines joining them are not joined automatically. A breakline that crosses another is skipped rather than split, and reported.
- **Existing ground plus the driveway layer.** There are no full cross-sections of the surface and no surface-to-surface earthworks volume. The cut and fill figure is indicative.
- **The driveway design is a screening aid.** It uses straight legs without vertical curves and does not check sight distance. Its defaults are the AS/NZS 2890.1:2004 domestic driveway figures only and do not include any council's own requirements beyond the kerb and layback shape entered.
- **The footpath check infers the footpath** from the design: it looks between the kerb and where the driveway first climbs steeper than the property-line limit.
- **The kerb face and layback legs are not checked.** This is deliberate.
- **The vehicle check uses the Figure C1 templates.** Real vehicles vary; enter other dimensions where a particular vehicle must be catered for. The scrape check and the transition-length rule test different things and can disagree.
- **Crossfall** is read between two alignments, not from a full cross-section of the surface. At a kerb the two faces may not line up square across.
- **File formats.** GradeLine reads DXF only and writes no DWG.
- **Desktop installer.** It is not code-signed, so Windows SmartScreen may warn on first run, and it does not update itself.

> **Important** GradeLine is an aid to the user's own checking, not a certified design tool. Final design, compliance assessment and sign-off belong in the user's normal design and approval workflow.

A. Default values

Setting	Default
Vertical exaggeration	1x (true scale)
True scale (1:1)	On
Existing terrain sample interval	2
Grade-break sensitivity (elevation tolerance)	0.05
Regular chainage marks / Datum line	Off (spacing 10) / Off
Ignore points with elevation (Z) = 0	On
Ignore points beyond a coordinate limit	Off (limit 0)
Use breaklines / Show breaklines on surface	On / On
Show triangulation edges / survey points / alignment	Off / Off / On
Menu sections / Theme	Collapsed / Dark
Maximum grade within the property	25%
First-leg (property line) limit	5%
Footpath check	On: 2.5% over at least 1.0 m
Transition threshold: summit / sag / minimum length	12.5% / 15% / 2.0 m
Layback template, road side / property side (rise / run)	25 / 450 mm; 70 / 450 mm
Design vehicle	B85, facing toward increasing chainage
Vehicle clearance / approach / departure	120 mm / 10° / 10°
Crossfall comparison method	Square across the alignments

B. Glossary

Term	Meaning
Alignment	A plan line along which the long section is cut.
Approach / departure angle	The angle from a tyre's ground contact to the lowest point of the front (approach) or rear (departure) overhang.
Breakline	A 3D line, such as a kerb or gutter string, that the surface's triangles are made to follow.
Chainage	Distance along an alignment from its start, in metres.
Crossfall	The difference in level across a driveway between two alignments, read square across them.
Datum line	A horizontal reference line at a chosen RL, drawn across the section.
Grade break	A point where the ground grade changes by more than the sensitivity tolerance.
Kerb invert / reveal	The gutter level at the kerb line / the vertical rise from the invert to the top of kerb.
Layback	The lowered kerb profile at a driveway crossover.
Long section	A profile of ground (or design) level plotted against chainage.
PVI	Point of vertical intersection: a design point where two straight grades meet.
RL	Reduced level: elevation relative to the survey datum.
Summit / sag	A grade change where the grade falls (summit, a crest) or rises (sag, a dip) in the direction of travel.
TIN	Triangulated irregular network: a surface built from triangles joining survey points.